

Small-time Controllability of Control Systems on 3D Lie Groups

Sam Xu

Department of Electrical and Computer Engineering
McGill University

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Problem Formulation

- We study control systems of the form

$$\dot{g} = (A + uB)g \quad (1)$$

where $g \in G$, $A, B \in \mathfrak{g}$, and u is the control

- G is a matrix Lie group, $\mathfrak{g}$ is the corresponding Lie algebra
- In particular, G is a 3D unimodular matrix Lie group, completely classified into $SU(2)$, $SL(2)$, H , $SE(2)$, $SH(2)$
- This project has two goals:
 1. Classify such systems on all 3D unimodular Lie groups
 2. Find all examples of such systems that are controllable in arbitrarily small time

Problem Formulation

Recall the definitions of these matrix Lie groups:

- $SU(2) = \left\{ \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} : a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1 \right\}$
- $SL(2) = \{X \in M_2(\mathbb{R}) : \det X = 1\}$
- $H = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$
- $SE(2) = \left\{ \begin{pmatrix} \cos \theta & -\sin \theta & x \\ \sin \theta & \cos \theta & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, \theta \in \mathbb{R} \right\}$
- $SH(2) = \left\{ \begin{pmatrix} \cosh \theta & \sinh \theta & x \\ \sinh \theta & \cosh \theta & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, \theta \in \mathbb{R} \right\}$

Problem Formulation

... and recall the basis elements for their corresponding Lie algebras:

- $\mathfrak{su}(2)$: $u_1 = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$; $u_2 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$; $u_3 = \frac{1}{2} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$
- $\mathfrak{sl}(2)$: $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$; $F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$; $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
- $\mathfrak{h}$: $X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$; $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$; $Z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
- $\mathfrak{sc}(2)$: $X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$; $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$; $Z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
- $\mathfrak{sh}(2)$: $X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$; $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$; $Z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Classifying Control Systems

- Notation: we denote (1) as (A, B)
- We want to simplify the system (A, B) into an equivalent system (A', B') , expressed cleanly using basis elements
- Our tools:
 - Conjugation: $(A, B) \sim (gAg^{-1}, gBg^{-1})$ for any $g \in G$
 - Control shifting $u = u' + \delta$: $(A, B) \sim ((A + \delta B), B)$
 - Control scaling $u = \mu u'$: $(A, B) \sim (A, \mu B)$
 - Time scaling by $1/\lambda$: $(A, B) \sim (\lambda A, \lambda B)$
- We do this for systems on each of the five matrix Lie groups
- There exists only finitely many such equivalent systems

Classification Example: $SU(2)$

First write A and B in terms of basis elements

$$A = a_1^0 u_1 + a_2^0 u_2 + a_3^0 u_3, \quad B = b_1^0 u_1 + b_2^0 u_2 + b_3^0 u_3$$

Conjugation group action acts as rotations in $\mathbb{R}^3$, preserving lengths and angles. Then,

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 u_1 + a_3^1 u_3, b_3^1 u_3) \\ &\sim (a_1^1 u_1, b_3^1 u_3) \\ &\sim (a_1^1 u_1, a_1^1 u_3) \\ &\sim (u_1, u_3)\end{aligned}$$

Hence $(A, B) \sim (u_1, u_3)$ on $SU(2)$.

Classification Results

$SU(2)$	(u_1, u_3)
$SL(2)$	(E, H) (F, H) $(\pm E \pm F, H)$ $(E, E - F)$ (H, E) (F, E)
H	(X, Z) (Z, X) (Y, X)
$SE(2)$	(Y, X) (Z, X) (X, Z)
$SH(2)$	(X, Z) (Y, Z) $(X + Y, Z)$ $(X, X + Y)$ $(Z, X + Y)$ (Z, Y) (X, Y) (Z, X) (Y, X)

Small-Time Controllability

- Small-time controllability (STC): the control system can be driven to any point in the space in arbitrarily small time
- Want to use Rashevsky-Chow Theorem to prove STC
- Problem: theorem only applies to driftless systems
- Strategy: use fast-oscillating controls and study the remaining drift
- First check if (A, B) is bracket generating, $\text{span}\{A, B, [A, B]\} = \mathfrak{g}$
- Then consider $\text{conv}(Ad_{e^{tB}}A)/\mathbb{R}B$. Search for new drift directions in this convexified adjoint orbit, and repeat the process
- End goal is to obtain an extended system, compatible with the original system, of the form

$$\dot{g} = (uX + vY + wZ)g$$

so that we can apply Rashevsky-Chow Theorem (but we might not reach this step)

Small-Time Controllability Example: $SU(2)$

System (1) on $SU(2)$ is

$$\dot{g} = (u_1 + uu_3)g$$

Bracket generating condition

$$[u_1, u_3] = -\frac{1}{2}u_2 \Rightarrow \text{span}\{u_1, u_3, [u_1, u_3]\} = \mathfrak{su}(2)$$

Proceed to compute the adjoint

$$Ad_{e^{\theta u_3}} u_1 = \{\cos(\theta)u_1 + \sin(\theta)u_2 : t \in \mathbb{R}\}$$

So

$$\text{conv}(Ad_{e^{\theta u_3}} u_1)/\mathbb{R}u_3 = \{(u_1, u_2) : u_1^2 + u_2^2 \leq 1\}$$

This is a bounded disc in the u_1 - u_2 plane. Hence (u_1, u_3) is not STC.

Small-Time Controllability Results

$SU(2)$	(u_1, u_3)	No
$SL(2)$	(E, H)	No
	(F, H)	No
	$(\pm E \pm F, H)$	No
	$(E, E - F)$	No
	(H, E)	No
	(F, E)	Yes
H	(X, Z)	No
	(Z, X)	No
	(Y, X)	No
$SE(2)$	(Y, X)	No
	(Z, X)	No
	(X, Z)	No
$SH(2)$	(X, Z)	No
	(Y, Z)	No
	$(X + Y, Z)$	No
	$(X, X + Y)$	No
	$(Z, X + Y)$	No
	(Z, Y)	No
	(X, Y)	No
	(Z, X)	No
	(Y, X)	No
		No

Conclusion

In this project,

1. I found equivalent systems for system (1) on each of $SU(2)$, $SL(2)$, H , $SE(2)$, $SH(2)$ - 22 in total
2. I determined the small-time controllability of each of the equivalent systems - only (F, E) in $SL(2)$ is small-time controllable